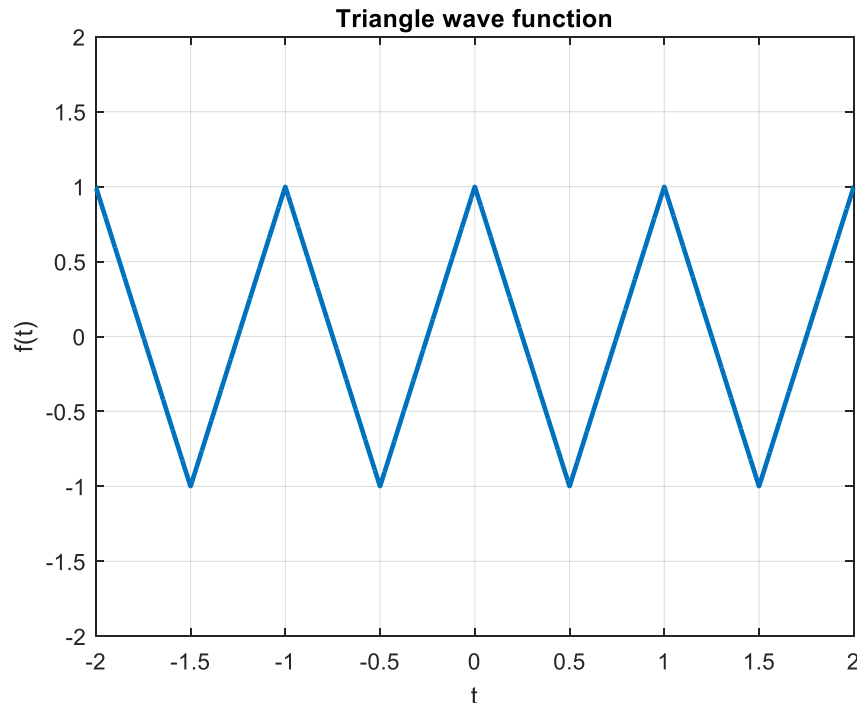


Fourier Transforms - Solution Sheet

- 1) Obtain the sinusoidal Fourier series expansion of the triangle function illustrated.

**SOLUTION**

First note that no particular time interval is specified - you can choose whatever full period you wish. If we choose the interval $[-1/2, 1/2]$, the resulting function has even symmetry over the interval and so a pure cosine Fourier series will be able to represent the result.

Also, the function would have odd half-wave symmetry, and so can be constructed from only odd harmonics.

Overall, this is an example of even quarter wave symmetry, where the function can be constructed from only odd cosine harmonics. so that we can write the function as

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t)$$

where all even terms in the cosine series will be zero

The next thing to note is that $a_0 = \frac{1}{T} \int_{\langle T \rangle} x(t) dt$ represents the average value of the function over the interval, which in this case is zero, hence

$$a_0 = 0$$

In general the remaining coefficients of a cosine series will be given by

$$a_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \cos(n\omega_0 t) dt$$

and in this specific case $\langle T \rangle = \left[-\frac{1}{2}, \frac{1}{2}\right]$, $T = 1$, and $\omega_0 = 2\pi$ so

$$a_n = 2 \left(\int_{-0.5}^0 (1 + 4t) \cos(2\pi n t) dt + \int_0^{0.5} (1 - 4t) \cos(2\pi n t) dt \right)$$

since in the first half of the period, the function is $x = (1 + 4t)$ and in the second half $x = (1 - 4t)$.

By symmetry, the first of these two integrals must be equal to the second, or use a change of variable $t = -t$ to show that

$$\begin{aligned} \int_{-0.5}^0 (1 + 4t) \cos(2\pi n t) dt &= - \int_{0.5}^0 (1 - 4\tau) \cos(-2\pi n \tau) d\tau \\ &= \int_0^{0.5} (1 - 4\tau) \cos(2\pi n \tau) d\tau \end{aligned}$$

Hence

$$\begin{aligned} a_n &= 4 \int_0^{0.5} (1 - 4t) \cos(2\pi n t) dt \\ &= 4 \int_0^{0.5} \cos(2\pi n t) dt - 16 \int_0^{0.5} t \cos(2\pi n t) dt \\ &= 4 \left[\frac{1}{2\pi n} \sin(2\pi n t) \right]_0^{0.5} - 16 \left(\left[\frac{t \sin(2\pi n t)}{2\pi n} \right]_0^{0.5} - \frac{1}{2\pi n} \int_0^{0.5} \sin(2\pi n t) dt \right) \end{aligned}$$

using integration by parts. Hence

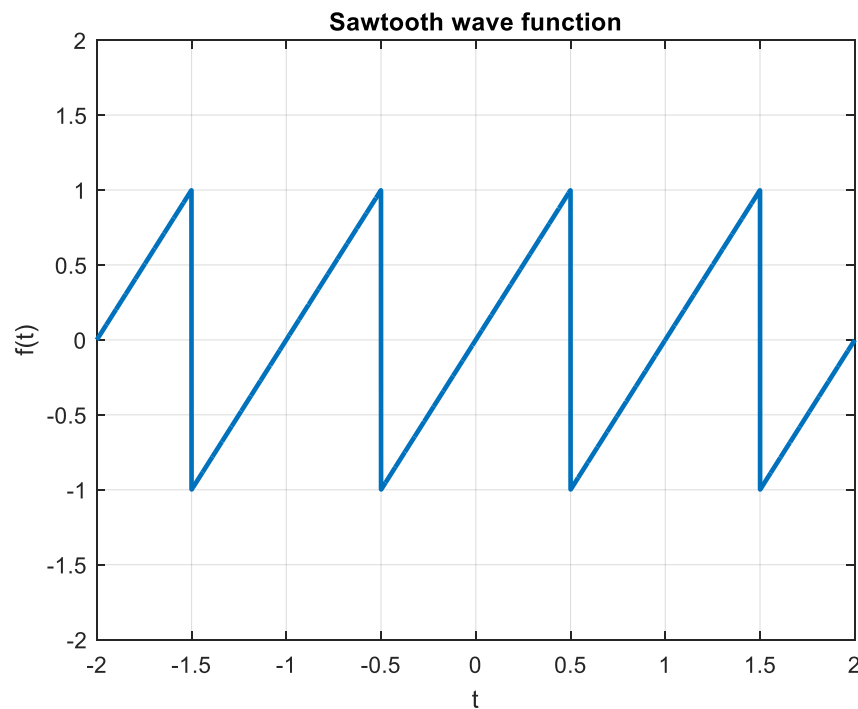
$$\begin{aligned} a_n &= 4 \left[\frac{1}{2\pi n} \sin(2\pi n t) \right]_0^{0.5} - 16 \left(\left[\frac{t \sin(2\pi n t)}{2\pi n} \right]_0^{0.5} + \frac{1}{(2\pi n)^2} [\cos(2\pi n t)]_0^{0.5} \right) \\ &= 0 - 16 \left(0 + \frac{1}{(2\pi n)^2} ((-1)^n - 1) \right) = \frac{4}{(\pi n)^2} (1 - (-1)^n) \end{aligned}$$

In full,

$$x(t) = \sum_{n=1}^{\infty} \frac{4}{(\pi n)^2} (1 - (-1)^n) \cos(2\pi n t)$$

As expected, all even terms in the cosine series are zero.

- 2) Obtain the sinusoidal Fourier series expansion of the sawtooth function illustrated.



SOLUTION

First note that no particular time interval is specified - you can choose whatever full period you wish. If we choose the interval $[-1/2, 1/2]$, the resulting function has odd symmetry over the interval and so a pure sine Fourier series will be able to represent the result so that we can write the function as

$$x(t) = \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

In general the coefficients of a sine series will be given by

$$b_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \sin(n\omega_0 t) dt$$

and in this specific case $\langle T \rangle = \left[-\frac{1}{2}, \frac{1}{2}\right]$, $T = 1$, and $\omega_0 = 2\pi$ so

$$\begin{aligned}
 b_n &= 2 \int_{-0.5}^{0.5} 2t \sin(2\pi nt) dt \\
 &= 4 \left(\left[-\frac{t \cos(2\pi nt)}{2\pi n} \right]_{-0.5}^{0.5} + \frac{1}{2\pi n} \int_{-0.5}^{0.5} \cos(2\pi nt) dt \right) \\
 &= 4 \left(\frac{1}{2\pi n} \left(-\frac{1}{2} \cos(n\pi) - \frac{1}{2} \cos(-n\pi) \right) + 0 \right) = \frac{2}{\pi n} (-1)^{n+1}
 \end{aligned}$$

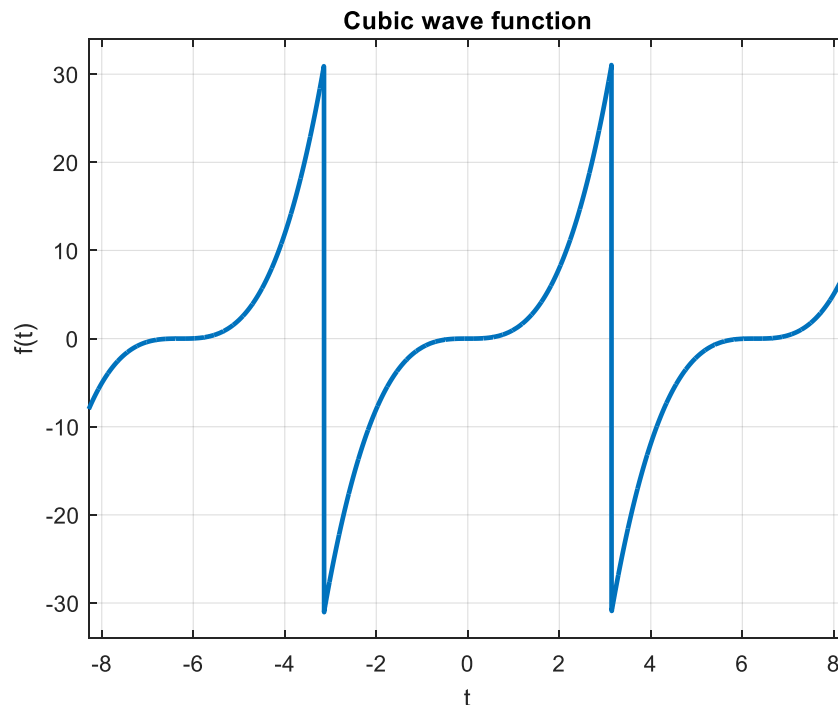
using integration by parts.

In full, the sinusoidal Fourier expansion is

$$x(t) = \sum_{n=1}^{\infty} \frac{2}{\pi n} (-1)^{n+1} \sin(2\pi nt)$$

- 3) Obtain the sinusoidal Fourier series expansion of the periodic function with period 2π illustrated and defined by

$$y = t^3 \text{ over the interval } -\pi < t < \pi$$



SOLUTION

It seems sensible to choose the interval $[-\pi, \pi]$. The resulting function has odd symmetry over the interval and so a pure sine Fourier series will be able to represent the result so that we can write the function as

$$x(t) = \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

In general the coefficients of a sine series will be given by

$$b_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \sin(n\omega_0 t) dt$$

and in this specific case $\langle T \rangle = [-\pi, \pi]$, $T = 2\pi$, and $\omega_0 = 1$ so

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} t^3 \sin(nt) dt \\ &= \frac{1}{\pi} \left(\left[-\frac{1}{n} t^3 \cos(nt) \right]_{-\pi}^{\pi} + \frac{1}{n} \int_{-\pi}^{\pi} 3t^2 \cos(nt) dt \right) \\ &= \frac{1}{\pi} \left(\left[-\frac{1}{n} t^3 \cos(nt) \right]_{-\pi}^{\pi} + \frac{1}{n} \left(\left[\frac{1}{n} 3t^2 \sin(nt) \right]_{-\pi}^{\pi} - \frac{1}{n} \int_{-\pi}^{\pi} 6t \sin(nt) dt \right) \right) \\ &= \frac{1}{\pi} \left(\left[-\frac{1}{n} t^3 \cos(nt) \right]_{-\pi}^{\pi} + \frac{1}{n} \left(\left[\frac{1}{n} 3t^2 \sin(nt) \right]_{-\pi}^{\pi} - \frac{1}{n} \left(\left[-\frac{1}{n} 6t \cos(nt) \right]_{-\pi}^{\pi} + \frac{1}{n} \int_{-\pi}^{\pi} 6 \cos(nt) dt \right) \right) \right) \\ &= \frac{1}{\pi} \left(\left[-\frac{1}{n} t^3 \cos(nt) \right]_{-\pi}^{\pi} + \frac{1}{n} \left(\left[\frac{1}{n} 3t^2 \sin(nt) \right]_{-\pi}^{\pi} - \frac{1}{n} \left(\left[-\frac{1}{n} 6t \cos(nt) \right]_{-\pi}^{\pi} + \frac{1}{n} \left[\frac{1}{n} 6 \sin(nt) \right]_{-\pi}^{\pi} \right) \right) \right) \\ &= \frac{1}{\pi n^4} [(6nt - n^3 t^3) \cos(nt) + (3n^2 t^2 - 6) \sin(nt)]_{-\pi}^{\pi} \\ &= \frac{2}{\pi n^4} (6n\pi - n^3 \pi^3)(-1)^n = \frac{2}{n^3} (6 - n^2 \pi^2)(-1)^n \end{aligned}$$

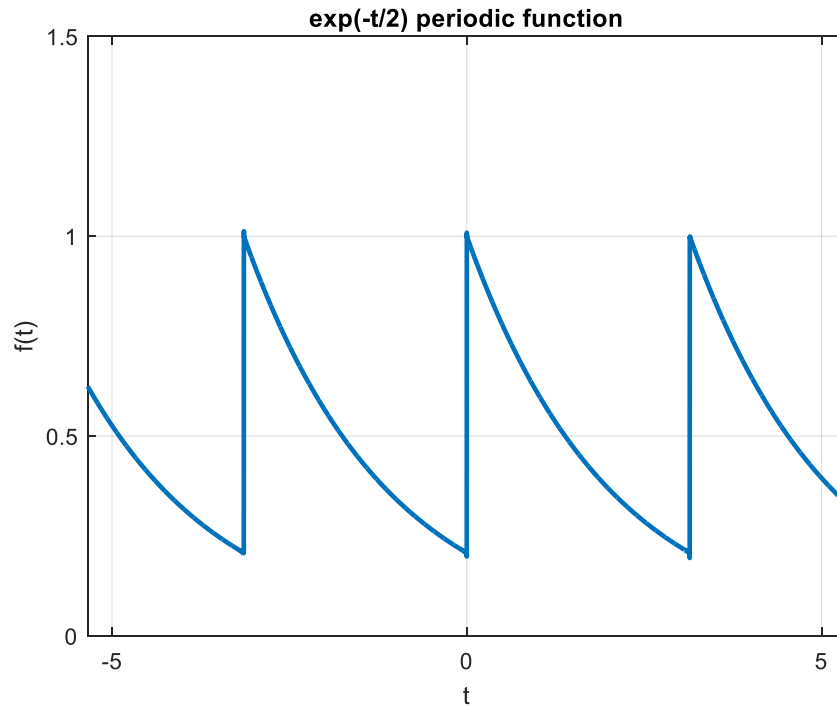
using integration by parts three times.

In full, the sinusoidal Fourier expansion is

$$x(t) = \sum_{n=1}^{\infty} \frac{2}{n^3} (6 - n^2 \pi^2)(-1)^n \sin(nt)$$

- 4) Obtain both the exponential and the sinusoidal Fourier series expansions of the periodic function with period π illustrated and defined by

$$y = e^{-0.5t} \text{ over the interval } 0 < t < \pi$$



SOLUTION

It seems sensible to choose the interval $[0, \pi]$. The resulting function has no particular symmetry over the interval and so a general Fourier series will be needed to represent the result - either in sinusoidal form:

$$x(t) = a_0 + \sum_n (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

where $\omega_0 = \frac{2\pi}{T}$ and the Fourier coefficients are given by

$$a_0 = \frac{1}{T} \int_{\langle T \rangle} x(t) dt$$

$$a_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \sin(n\omega_0 t) dt$$

or in complex exponential form:

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

where $c_0 = a_0$ and $c_n = \frac{1}{T} \int_{\langle T \rangle} x(t) \exp(-jn\omega_0 t) dt$

The expansions are related by

$$c_n = \frac{1}{2} (a_n - jb_n) \text{ and } c_{-n} = \frac{1}{2} (a_n + jb_n) = \overline{c_n} \text{ for positive } n$$

and correspondingly $a_n = (c_n + c_{-n})$ and $b_n = j(c_n - c_{-n})$

In this specific case $\langle T \rangle = [0, \pi]$, $T = \pi$, and $\omega_0 = 2$ so that starting with the complex series expansion,

$$c_0 = \frac{1}{\pi} \int_0^\pi e^{-\frac{t}{2}} dt = \frac{1}{\pi} \left[-2 e^{-\frac{t}{2}} \right]_0^\pi = \frac{2}{\pi} \left(1 - e^{-\frac{\pi}{2}} \right)$$

and

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^\pi e^{-\frac{t}{2}} e^{-2jnt} dt = \frac{1}{\pi} \int_0^\pi e^{-\left(\frac{t}{2} + 2jnt\right)} dt \\ &= \frac{1}{\pi} \left[-\frac{1}{\frac{1}{2} + 2jn} e^{-\left(\frac{t}{2} + 2jnt\right)} \right]_0^\pi = -\frac{2}{\pi(1 + 4jn)} \left(e^{-\left(\frac{\pi}{2} + 2jn\pi\right)} - 1 \right) = \frac{2}{\pi(1 + 4jn)} \left(1 - e^{-\frac{\pi}{2}} \right) \end{aligned}$$

Hence, in full, the complex exponential Fourier series expansion is

$$x(t) = \frac{2}{\pi} \left(1 - e^{-\frac{\pi}{2}} \right) \sum_{n=-\infty}^{\infty} \frac{1}{1 + 4jn} e^{jn\omega_0 t}$$

The corresponding sinusoidal coefficients are

$$\begin{aligned} a_n &= (c_n + c_{-n}) = \frac{2}{\pi} \left(1 - e^{-\frac{\pi}{2}} \right) \left(\frac{1}{1 + 4nj} + \frac{1}{1 - 4nj} \right) \\ &= \frac{2}{\pi} \left(1 - e^{-\frac{\pi}{2}} \right) \left(\frac{2}{1 + 16n^2} \right) = \frac{4}{(1 + 16n^2)\pi} \left(1 - e^{-\frac{\pi}{2}} \right) \end{aligned}$$

and

$$\begin{aligned} b_n &= j(c_n - c_{-n}) = \frac{2j}{\pi} \left(1 - e^{-\frac{\pi}{2}} \right) \left(\frac{1}{1 + 4nj} - \frac{1}{1 - 4nj} \right) \\ &= \frac{2j}{\pi} \left(1 - e^{-\frac{\pi}{2}} \right) \left(-\frac{8nj}{1 + 16n^2} \right) = \frac{16n}{(1 + 16n^2)\pi} \left(1 - e^{-\frac{\pi}{2}} \right) \end{aligned}$$

- 5) For questions (1) - (4) evaluate the total energy and average power of each of the signals over one complete period.

SOLUTION

In general the total energy over a single period of a periodic signal is

$$E = \int_{\langle T \rangle} |x(t)|^2 dt$$

and the average power over a single period is

$$P = \frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt$$

or, if viewed in the complex frequency domain, by Parseval's Theorem the average power can also be written as

$$P = \sum_{n=-\infty}^{\infty} |c_n|^2$$

For the signals in question

(i) For the triangle function in Q1, since the function has a discontinuity in gradient, the integral needs to be split into two parts

$$E = \int_{-0.5}^0 (1 + 4t)^2 dt + \int_0^{0.5} (1 - 4t)^2 dt$$

and by symmetry (or by a coordinate transformation) these two integrals are equal, so that the total energy over one period is

$$\begin{aligned} E &= 2 \int_0^{0.5} (1 - 4t)^2 dt = 32 \int_0^{0.5} \left(\frac{1}{4} - t\right)^2 dt \\ &= 32 \left[-\frac{1}{3} \left(\frac{1}{4} - t\right)^3 \right]_0^{0.5} = -\frac{32}{3} \left[\left(-\frac{1}{4}\right)^3 - \left(\frac{1}{4}\right)^3 \right]_0^{0.5} = \frac{1}{3} \end{aligned}$$

and the average power is

$$P = \frac{E}{T} = \frac{\frac{1}{3}}{1} = \frac{1}{3}$$

(ii) For the sawtooth function in Q2, the energy integral over one period is

$$E = \int_{-0.5}^{0.5} (2t)^2 dt = \left[\frac{4}{3} t^3 \right]_{-0.5}^{0.5} = \frac{4}{3} \left(\frac{1}{8} - -\frac{1}{8} \right) = \frac{1}{3}$$

and the average power is

$$P = \frac{E}{T} = \frac{\frac{1}{3}}{1} = \frac{1}{3}$$

(iii) For the cubic function in Q3, the energy integral over one period is

$$E = \int_{-\pi}^{\pi} (t^3)^2 dt = \left[\frac{1}{7} t^7 \right]_{-\pi}^{\pi} = \frac{1}{7} (\pi^7 - -\pi^7) = \frac{2}{7} \pi^7$$

and the average power is

$$P = \frac{E}{T} = \frac{1}{2\pi} \left(\frac{2}{7} \pi^7 \right) = \frac{1}{7} \pi^6$$

(iv) For the periodic exponential function in Q4, the energy integral over one period is

$$E = \int_0^{\pi} (e^{-0.5t})^2 dt = [-e^{-t}]_0^{\pi} = 1 - e^{-\pi}$$

and the average power is

$$P = \frac{1}{\pi} (1 - e^{-\pi})$$

- 6) A periodic signal with a repeat period of three milliseconds exhibits half-wave symmetry. You are required to purchase an amplifier system to process the signal, where the primary aim is to amplify as many harmonics of the signal as possible, and then to achieve this at the lowest cost.

You have the option of five systems:

- A) Bandwidth 2.5 kHz, costing £300;
- B) Bandwidth 2.7 kHz, costing £350;
- C) Bandwidth 2.9 kHz, costing £400;
- D) Bandwidth 3.1 kHz, costing £450;
- E) Bandwidth 3.3 kHz, costing £500.

Which would be the most appropriate system to purchase - and why?

SOLUTION

Since the signal is periodic, it can be represented in frequency form as a Fourier series with fundamental frequency $f_0 = \frac{1}{3 \times 10^{-3}} = \frac{1}{3} \times 10^3 \text{ Hz} = \frac{1}{3} \text{ kHz}$. Also, since it exhibits half-wave symmetry, it can be constructed from only odd harmonics. Hence the harmonics of the periodic signal will occur at the specific frequencies:

$$\begin{aligned} f_0 &= \frac{1}{3} \text{ kHz} \\ 3f_0 &= 1 \text{ kHz} \\ 5f_0 &= \frac{5}{3} \text{ kHz} \\ 7f_0 &= \frac{7}{3} \text{ kHz} \\ 9f_0 &= 3 \text{ kHz} \\ 11f_0 &= \frac{11}{3} \text{ kHz} \\ &\vdots \\ &\text{etc.} \end{aligned}$$

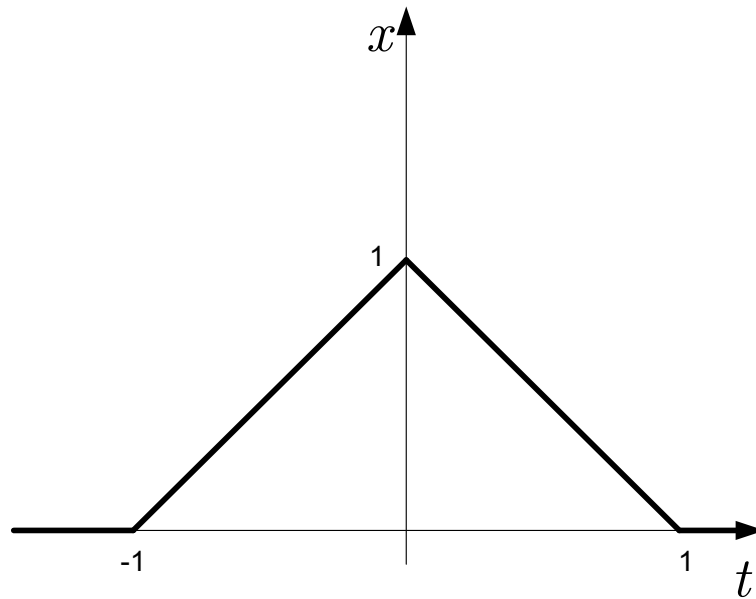
None of the suggested amplifiers has adequate bandwidth to handle the 11th harmonic, but both systems (D) and (E) can process up to and including the 9th harmonic at 3kHz. The optimal choice would be system (D), since the additional bandwidth provided by the more expensive system is not sufficient to process any more harmonics.

7) Sketch the triangular pulse function defined by

$$x(t) = \begin{cases} t + 1, & \text{for } -1 < t \leq 0 \\ -t + 1, & \text{for } 0 < t < 1 \\ 0, & \text{else} \end{cases}$$

Evaluate the total energy of the function and its Fourier transform.

SOLUTION



The total energy of the signal in general is

$$E_{Total} = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

in time domain form. Hence for this signal

$$\begin{aligned} E_{Total} &= \int_{-1}^0 (1+t)^2 dt + \int_0^1 (1-t)^2 dt \\ &= 2 \int_0^1 (1-t)^2 dt = 2 \left[-\frac{1}{3} (1-t)^3 \right]_0^1 = 2 \left(-0 + \frac{1}{3} \right) = \frac{2}{3} \end{aligned}$$

using the fact that the signal is symmetric.

The Fourier transform is defined in general as

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt$$

so here

$$X(\omega) = \int_{-1}^0 (1+t) \exp(-j\omega t) dt + \int_0^1 (1-t) \exp(-j\omega t) dt$$

Use a change of variable $\tau = -t$ in the first integral to show that

$$\begin{aligned}\int_{-1}^0 (1+t) \exp(-j\omega t) dt &= -\int_1^0 (1-\tau) \exp(j\omega\tau) d\tau \\ &= \int_0^1 (1-\tau) \exp(j\omega\tau) d\tau = \int_0^1 (1-t) \exp(j\omega t) dt\end{aligned}$$

so that

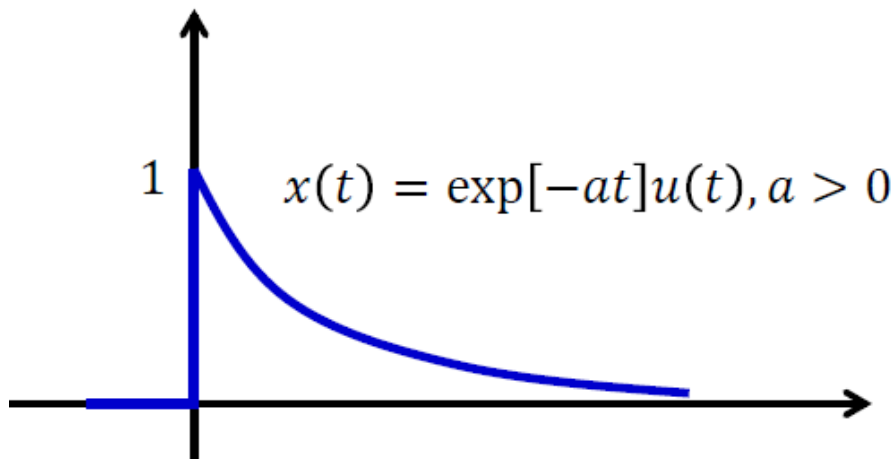
$$\begin{aligned}X(\omega) &= \int_0^1 (1-t) \exp(j\omega t) dt + \int_0^1 (1-t) \exp(-j\omega t) dt \\ &= \int_0^1 (1-t)(\exp(j\omega t) + \exp(-j\omega t)) dt \\ &= 2 \int_0^1 (1-t) \cos(\omega t) dt \\ &= 2 \left(\left[\frac{1}{\omega} (1-t) \sin(\omega t) \right]_0^1 + \frac{1}{\omega} \int_0^1 \sin(\omega t) dt \right) \\ &= 2 \left(\left[\frac{1}{\omega} (1-t) \sin(\omega t) \right]_0^1 + \frac{1}{\omega} \left[-\frac{1}{\omega} \cos(\omega t) \right]_0^1 \right) \\ &= 2 \left(0 - 0 - \frac{1}{\omega^2} (\cos(\omega) - 1) \right) \\ &= \frac{2}{\omega^2} (1 - \cos(\omega)) = \frac{4}{\omega^2} \sin^2 \left(\frac{\omega}{2} \right) = \text{sinc}^2 \left(\frac{\omega}{2} \right)\end{aligned}$$

8) Sketch the function defined by

$$x(t) = e^{-at} u(t) \text{ where } a > 0$$

Evaluate the total energy of the function and its Fourier transform.

SOLUTION



The total energy of the signal in general is $E_{Total} = \int_{-\infty}^{\infty} |x(t)|^2 dt$ in time domain form. Hence for this signal

$$E_{Total} = \int_0^{\infty} (e^{-at})^2 dt = \int_0^{\infty} e^{-2at} dt = \left[-\frac{1}{2a} e^{-2at} \right]_0^{\infty} = \frac{1}{2a}$$

The Fourier transform is defined in general as

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt$$

so here

$$\begin{aligned} X(\omega) &= \int_0^{\infty} \exp(-at) \exp(-j\omega t) dt \\ &= \int_0^{\infty} \exp(-(a + j\omega)t) dt = \left[-\frac{1}{a + j\omega} \exp(-(a + j\omega)t) \right]_0^{\infty} = \frac{1}{a + j\omega} \end{aligned}$$

assuming convergence as $t \rightarrow \infty$.

NOTE: Knowing the Fourier transform of the signal, it is now possible to use Parseval's theorem to establish the total energy another way:

$$E_{Total} = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

So, since

$$\begin{aligned} |X(\omega)|^2 &= \left(\frac{1}{a + j\omega} \right) \left(\frac{1}{a - j\omega} \right) = \frac{1}{a^2 + \omega^2} \\ E_{Total} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{a^2 + \omega^2} d\omega = \frac{1}{2\pi} \left[\frac{1}{a} \arctan \left(\frac{\omega}{a} \right) \right]_{-\infty}^{\infty} = \frac{1}{2\pi a} \left(\frac{\pi}{2} - -\frac{\pi}{2} \right) = \frac{1}{2a} \end{aligned}$$

9) In the Multivariable Tools notes we saw (p50-51) that

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Hence evaluate the total energy and the Fourier transform $G(\omega)$ of the Gaussian function $g(t) = \exp(-at^2)$.

What is the full width at half maximum (FWHM) value for the spectral peak defined by $G(\omega)$?

SOLUTION

The total energy of the signal in general is

$$E_{Total} = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

in time domain form. Hence for this signal

$$E_{Total} = \int_{-\infty}^{\infty} (e^{-at^2})^2 dt = 2 \int_0^{\infty} (e^{-at^2})^2 dt = 2 \int_0^{\infty} e^{-2at^2} dt$$

Make a change of variable $\tau = \sqrt{2a}t$, so that $d\tau = \sqrt{2a} dt$, so that

$$E_{Total} = 2 \int_0^{\infty} e^{-2at^2} dt = 2 \frac{1}{\sqrt{2a}} \int_0^{\infty} e^{-\tau^2} d\tau = 2 \frac{1}{\sqrt{2a}} \frac{\sqrt{\pi}}{2} = \sqrt{\frac{\pi}{2a}}$$

The Fourier transform is defined in general as

$$G(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt$$

so here

$$G(\omega) = \int_{-\infty}^{\infty} \exp(-at^2) \exp(-j\omega t) dt$$

This integral is awkward. One approach is to differentiate both sides with respect to ω , giving

$$\frac{d}{d\omega} G(\omega) = \int_{-\infty}^{\infty} (-jt) \exp(-at^2) \exp(-j\omega t) dt$$

and then use integration by parts, setting

$$u = \exp(-j\omega t) \quad \text{fi} \quad du = -j\omega \exp(-j\omega t)$$

$$\text{and} \quad dv = (-t) \exp(-at^2) \quad \text{fi} \quad v = \frac{1}{2a} \exp(-at^2)$$

Hence

$$\begin{aligned} \frac{d}{d\omega} G(\omega) &= \int_{-\infty}^{\infty} (-jt) \exp(-at^2) \exp(-j\omega t) dt \\ &= j \left[\exp(-j\omega t) \frac{1}{2a} \exp(-at^2) \right]_{-\infty}^{\infty} - j(-j\omega) \frac{1}{2a} \int_{-\infty}^{\infty} \exp(-at^2) \exp(-j\omega t) dt \\ &= 0 - 0 - \frac{\omega}{2a} G(\omega) \end{aligned}$$

This is a differential equation which says that the Fourier transform must satisfy:

$$\frac{dG}{d\omega}(\omega) = -\frac{\omega}{2a} G(\omega)$$

This is just a variables separable form:

$$\begin{aligned} \frac{d}{d\omega} G(\omega) &= \frac{\omega}{2a} G(\omega) \Rightarrow \int \frac{1}{G} dG = -\frac{1}{2a} \int \omega d\omega \\ &\Rightarrow \ln(G) = -\frac{1}{4a} \omega^2 + K \\ &\Rightarrow G(\omega) = A \exp\left(-\frac{\omega^2}{4a}\right) \end{aligned}$$

where K and A are constants of integration. To establish A , note that $A = X(0)$, which is just the average value of the time domain signal:

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = 2 \int_0^{\infty} e^{-ax^2} dx = 2 \frac{1}{\sqrt{a}} \int_0^{\infty} e^{-\tau^2} d\tau = 2 \frac{1}{\sqrt{a}} \frac{\sqrt{\pi}}{2} = \sqrt{\frac{\pi}{a}}$$

in a similar fashion to before, but now using a change of variable $\tau = \sqrt{a} t$, so that $d\tau = \sqrt{a} dt$. Hence, overall,

if
$$g(t) = \exp(-at^2)$$

then
$$G(\omega) = \sqrt{\frac{\pi}{a}} \exp\left(-\frac{\omega^2}{4a}\right)$$

NOTE This is a crucial result - that the Fourier transform of a Gaussian pulse is itself a Gaussian pulse.

The maximum value of the frequency domain pulse occurs when $\omega = 0$ and is $G(0) = \sqrt{\frac{\pi}{a}}$

The full width at half maximum (FWHM) is given by the frequencies when $G(\omega) = \frac{1}{2} \sqrt{\frac{\pi}{a}}$, so we need to solve

$$\sqrt{\frac{\pi}{a}} \exp\left(-\frac{\omega^2}{4a}\right) = \frac{1}{2} \sqrt{\frac{\pi}{a}}$$

for ω .

$$\begin{aligned} \exp\left(-\frac{\omega^2}{4a}\right) &= \frac{1}{2} \rightarrow -\frac{\omega^2}{4a} = \ln\left(\frac{1}{2}\right) \\ \rightarrow \omega &= \pm 2 \sqrt{a \ln(2)} \end{aligned}$$

The FWHM is hence $4\sqrt{a \ln(2)}$.

10) Evaluate the Fourier transform of the function $x(t) = e^{-at} \sin(\omega_0 t) u(t)$

SOLUTION

This can be evaluated from first principles from the general Fourier transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt$$

using the exponential form of the sine function

$$\sin(\omega_0 t) = \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t})$$

and the result from Q8 that

$$\Im[e^{-at} u(t)] = \frac{1}{a + j\omega}$$

Hence

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} e^{-at} \sin(\omega_0 t) u(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} e^{-at} \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t}) e^{-j\omega t} dt \\ &= \frac{1}{2j} \int_0^{\infty} e^{-at} (e^{j\omega_0 t} - e^{-j\omega_0 t}) e^{-j\omega t} dt \\ &= \frac{1}{2j} \left(\int_0^{\infty} e^{-(a-j\omega_0)t} e^{-j\omega t} dt - \int_0^{\infty} e^{-(a+j\omega_0)t} e^{-j\omega t} dt \right) \\ &= \frac{1}{2j} \left(\frac{1}{a-j\omega_0+j\omega} - \frac{1}{a+j\omega_0+j\omega} \right) \\ &= \frac{1}{2j} \left(\frac{2j\omega_0}{a^2 + \omega_0^2 - \omega^2 + 2j\omega a} \right) = \frac{\omega_0}{a^2 + \omega_0^2 - \omega^2 + 2j\omega a} \end{aligned}$$

PAST EXAMINATION QUESTIONS

(2019 Stage 2 Examinations, question 5)

- (a) Evaluate the sinusoidal Fourier series expansion of the periodic function $x(t) = 2t + 1$ defined over the time interval $-3 \leq t \leq 3$

[12 marks]

- (b) For a continuous-time signal $x(t)$, state the conditions under which it is possible to evaluate the Fourier transform of the signal.

Confirm that the two-sided decaying exponential signal

$$x(t) = e^{-a|t|}$$

satisfies these conditions and evaluate

$$X(\omega) = \mathcal{F}[e^{-a|t|}]$$

What is the full width at half maximum (FWHM) value for the spectral peak defined by $X(\omega)$?

[13 marks]

SOLUTION

- (a) In general for a periodic function of period T , the Fourier series expansion is given by

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

where $\omega_0 = \frac{2\pi}{T}$ and the Fourier coefficients are given by

$$a_0 = \frac{1}{T} \int_{\langle T \rangle} x(t) dt$$

$$a_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \sin(n\omega_0 t) dt$$

(Mostly directly from the supplied formula sheet)

In this particular case $\omega_0 = \frac{2\pi}{6} = \frac{\pi}{3}$ and

$$a_0 = \frac{1}{6} \int_{-3}^3 (2t + 1) dt = \frac{1}{6} [t^2 + t]_{-3}^3 = \frac{1}{6} (12 - 6) = 1$$

$$a_n = \frac{1}{3} \int_{-3}^3 (2t + 1) \cos\left(\frac{n\pi t}{3}\right) dt$$

$$\begin{aligned}
&= \frac{1}{3} \left(\left[\frac{3}{n\pi} (2t+1) \sin\left(\frac{n\pi t}{3}\right) \right]_{-3}^3 - \frac{3}{n\pi} \int_{-3}^3 2 \sin\left(\frac{n\pi t}{3}\right) dt \right) \\
&= \frac{1}{3} \left(\left[\frac{3}{n\pi} (2t+1) \sin\left(\frac{n\pi t}{3}\right) \right]_{-3}^3 - \frac{3}{n\pi} \left[-\frac{3}{n\pi} 2 \cos\left(\frac{n\pi t}{3}\right) \right]_{-3}^3 \right) \\
&= (0 - 0) - (0 - 0) = 0
\end{aligned}$$

using integration by parts. Similarly

$$\begin{aligned}
b_n &= \frac{2}{T} \int_{-3}^3 (2t+1) \sin\left(\frac{n\pi t}{3}\right) dt \\
&= \frac{1}{3} \left(\left[-\frac{3}{n\pi} (2t+1) \cos\left(\frac{n\pi t}{3}\right) \right]_{-3}^3 + \frac{3}{n\pi} \int_{-3}^3 2 \cos\left(\frac{n\pi t}{3}\right) dt \right) \\
&= \frac{1}{3} \left(\left[-\frac{3}{n\pi} (2t+1) \cos\left(\frac{n\pi t}{3}\right) \right]_{-3}^3 + \frac{3}{n\pi} \left[\frac{3}{n\pi} 2 \sin\left(\frac{n\pi t}{3}\right) \right]_{-3}^3 \right) \\
&= -\frac{1}{n\pi} \left((7 - (-5)) \cos(n\pi) - (0 - 0) \right) = -\frac{12}{n\pi} \cos(n\pi)
\end{aligned}$$

So that the Fourier series representation is:

$$x(t) = 1 - \frac{12}{\pi} \sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n} \sin\left(\frac{n\pi t}{3}\right)$$

or, noting that $\cos(n\pi) = (-1)^n$

$$x(t) = 1 - \frac{12}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin\left(\frac{n\pi t}{3}\right)$$

(b) The signal must be aperiodic and have finite energy.

For the two-sided decaying exponential signal

$$x(t) = e^{-a|t|}$$

where $a > 0$, the function does not display any periodicity and the total energy is

$$E = \int_{-\infty}^{\infty} e^{-a|t|} dt$$

$$\begin{aligned}
&= \int_{-\infty}^0 e^{at} dt + \int_0^{\infty} e^{-at} dt \\
&= \frac{1}{a} [e^{at}]_{-\infty}^0 - \frac{1}{a} [e^{-at}]_0^{\infty} \\
&= \frac{1}{a} (1 - 0) - \frac{1}{a} (0 - 1) = \frac{2}{a}
\end{aligned}$$

and so is finite.

The Fourier transform is

$$\begin{aligned}
X(\omega) &= \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} dt \\
&= \int_{-\infty}^0 e^{at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt \\
&= \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt \\
&= \frac{1}{a-j\omega} [e^{(a-j\omega)t}]_{-\infty}^0 - \frac{1}{a+j\omega} [e^{-(a+j\omega)t}]_0^{\infty} \\
&= \frac{1}{a-j\omega} (1 - 0) - \frac{1}{a+j\omega} (0 - 1) \\
&= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} = \frac{2a}{a^2 + \omega^2}
\end{aligned}$$

The peak value of $X(\omega)$ occurs at $\omega = 0$ and is $X(0) = \frac{2}{a}$.

The FWHM value occurs when $X(\omega)$ is half of this peak value: $\frac{2a}{a^2 + \omega^2} = \frac{1}{a}$

so

$$2a^2 = a^2 + \omega^2$$

$$\Rightarrow \omega^2 = a^2$$

$$\Rightarrow \omega = a$$

(2019 Stage 2 Examinations, question 6)

- (a) Explain the concept of odd half-wave symmetry, explaining the type of signal to which it can be applied. Draw an example of an even function which displays odd half-wave symmetry.

- (i) The first half of a periodic signal, $x(t)$, is shown in Figure 6(a), over the time interval $-\frac{T}{2} < t \leq 0$.

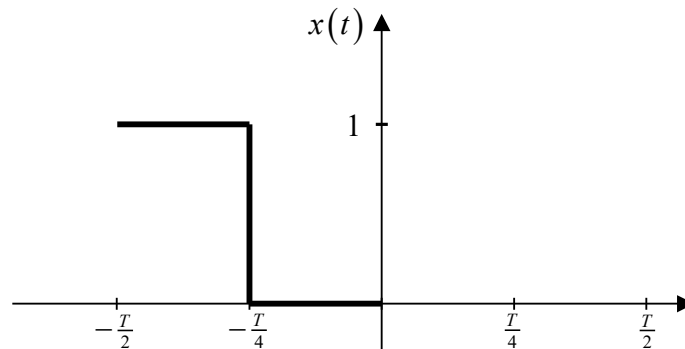


Figure 6(a)

The signal has odd half-wave symmetry. Copy Figure 6(a) into your answer book and complete the sketch of $x(t)$ for two periods.

- (ii) State which coefficients will be zero in the exponential Fourier series representation for $x(t)$.

- (iii) You are required to purchase an amplifier system to process a signal similar to $x(t)$ which has a period of $T = 0.5$ ms (milliseconds). The primary aim is to amplify as many harmonics of the signal as possible, and then to achieve this at the lowest cost. You have the option of three systems:

- A) Bandwidth 15 kHz, cost £300;
- B) Bandwidth 16 kHz, cost £500;
- C) Bandwidth 17 kHz, cost £700.

Which would you choose and why?

[12 marks]

- (b)

- (i) The autocorrelation function $r_{xx}(t)$ for a real, continuous-time, time-limited deterministic signal $x(t)$ observed over a period $[-\frac{T}{2}, \frac{T}{2}]$ is defined as

$$r_{xx}(t) = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) x(\tau + t) d\tau$$

State and prove the autocorrelation (or Wiener–Khinchin) theorem.

(ii) Sketch a plot showing the autocorrelation function of a single rectangular pulse of unit height and width T .

[13 marks]

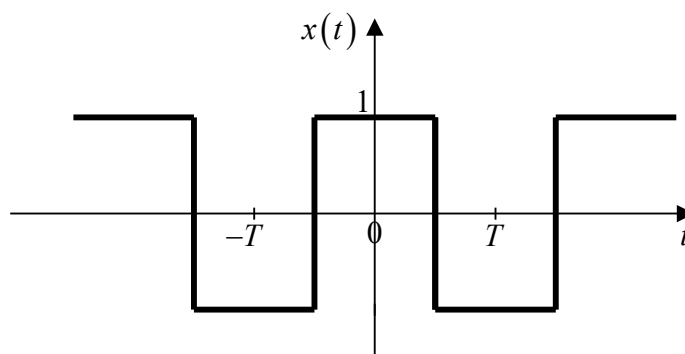
SOLUTION

(a) The concept of odd half-wave symmetry is applicable to PERIODIC signals and describes the case where, when examining a SINGLE PERIOD (of length T), of the signal the values of second half of the signal period are merely the NEGATIVE of those of the first half. In detail:

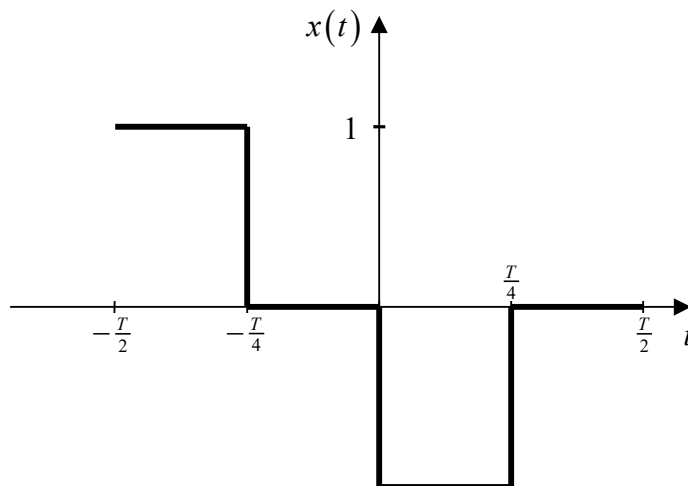
$$f(t) = -f\left(t + \frac{T}{2}\right)$$

so that shifting the second half of the signal through one-half of a period and adding it to the first half will produce a zero result.

A simple example of an even function which displays odd half-wave symmetry could be



(i) The required sketch is:



(ii) For odd half wave symmetry only odd harmonics are required to describe the function. Thus ALL even harmonics in the exponential Fourier series will be zero:

$$c_{2n} = 0 \quad n = \pm 1, n = \pm 2, n = \pm 3 \dots$$

(iii) The fundamental frequency is $\frac{1}{T}$, or 2 kHz in this specific case. Since we only need odd

harmonics in the Fourier series the frequencies in the Fourier series are:

1st: 2 kHz

3rd: 6 kHz

5th: 10 kHz

7th: 14 kHz

9th: 18 kHz

Looking at the frequency ranges available, all three amplifiers can operate at the fundamental and 3rd, 5th and 7th harmonics, but none go up to the 9th harmonic. So the best option is option A, as it is the cheapest which covers the frequency range up to the 7th harmonic.

(b)

(i) The autocorrelation (or Wiener-Khinchin) theorem states that the Fourier transform of the autocorrelation function is the Power Spectral Density:

$$\mathcal{F}[r_{xx}(t)] = \mathcal{F}\left[\frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) x(\tau + t) d\tau\right] = \frac{|X(\omega)|^2}{T}$$

By definition, the Fourier transform of the autocorrelation function is

$$\mathcal{F}[r_{xx}(t)] = \mathcal{F}\left[\frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) x(\tau + t) d\tau\right] = \int_{-\infty}^{\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) x(\tau + t) d\tau e^{-j\omega t} dt$$

and reversing the order of integration gives

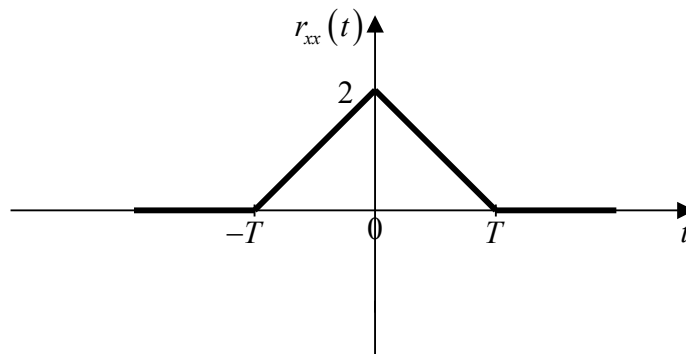
$$\begin{aligned} \mathcal{F}[r_{xx}(t)] &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \int_{-\infty}^{\infty} x(\tau) x(\tau + t) e^{-j\omega t} dt d\tau \\ &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) \left[\int_{-\infty}^{\infty} x(\tau + t) e^{-j\omega t} dt \right] d\tau \end{aligned}$$

and finally, by using the substitution $u = \tau + t$:

$$\begin{aligned} \mathcal{F}[r_{xx}(t)] &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) \left[\int_{-\infty}^{\infty} x(u) e^{-j\omega(u-\tau)} du \right] d\tau \\ &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) e^{j\omega\tau} \left[\int_{-\infty}^{\infty} x(u) e^{-j\omega u} du \right] d\tau \\ &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) e^{j\omega\tau} [X(\omega)] d\tau \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{T} X(\omega) \int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) e^{j\omega\tau} d\tau \\
&= \frac{1}{T} X(\omega) \left[\int_{-\frac{T}{2}}^{\frac{T}{2}} x(\tau) e^{-j\omega\tau} d\tau \right]^* = \frac{1}{T} X(\omega) X^*(\omega) = \frac{|X(\omega)|^2}{T}
\end{aligned}$$

The autocorrelation function of a single rectangular pulse of unit height and width T looks like



(2020 Stage 2 Examinations, question 5)

(a)

[10 Marks]

Determine the Fourier transform $X(\omega)$ of a rectangular pulse $x(t)$ which is given as

$$x(t) = \begin{cases} -1, & -2 \leq t < 0 \\ 3, & 4 \leq t < 6 \\ 0, & \text{otherwise} \end{cases}$$

(b)

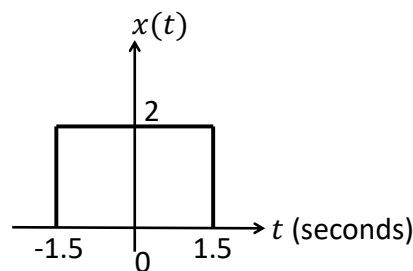
[7 Marks]

The signal defined in part 5(a) is convolved with the signal $h(t)$, according to the equation $z(t) = x(t) * h(t)$, where $h(t) = 3\delta(t - 2) + \delta(t - 4)$. Derive an expression for the Fourier Transform $Z(\omega)$ of the signal $z(t)$.

(c)

[8 Marks]

Let a signal $x(t)$ illustrated below be modulated by $c(t) = \cos(\omega_0 t)$. Determine mathematically its Fourier Transform, referring to Fourier Transform of a cosine signal.



SOLUTION

(a)

We can rewrite $x(t)$ as a superposition of two signals in the following:

$$x(t) = x_1(t) + x_2(t),$$

Where

$$x_1(t) = -\text{rect}\left(\frac{t+1}{2}\right), \quad x_2(t) = 3\text{rect}\left(\frac{t-5}{2}\right) \text{ and}$$

$$\text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1, & |t| \leq T/2 \\ 0, & |t| > T/2 \end{cases}$$

Where $T = 2$.

Here, $x_1(t)$ and $x_2(t)$ are a scaled and time-shifted version of $rect(t)$.

Note: $F\left[rect\left(\frac{t}{T}\right)\right] = T \text{sinc}\left(\frac{\omega T}{2}\right)$ and

the Second-shifting theorem: $F[x(t - \tau)] = e^{-j\omega\tau}X(\omega)$

Applying these to $F[x(t)]$, we have

$$X_1(\omega) = F[x_1(t)] = -e^{j\omega} 2 \text{sinc}(\omega)$$

$$X_2(\omega) = F[x_2(t)] = 3e^{-j5\omega} 2 \text{sinc}(\omega)$$

Accordingly, due to the linearity property of the Fourier Transform,

$$X(\omega) = X_1(\omega) + X_2(\omega) = -e^{-j2\omega} 2 \text{sinc}(\omega) (e^{j3\omega} - 3e^{-j3\omega})$$

(b)

$$h(t) = 3\delta(t - 2) + \delta(t - 4)$$

$$x(t) = -rect\left(\frac{t+1}{2}\right) + 3rect\left(\frac{t-5}{2}\right)$$

Note: $z(t) = h(t) * x(t) \rightarrow Z(\omega) = H(\omega)X(\omega)$.

Using the FT pairs, i.e., $F(\delta(t)) = 1$ and the Second-shifting theorem,

$$H(\omega) = F[h(t)] = 3e^{-j\omega 2} + e^{-j\omega 4}$$

$$Z(\omega) = [3e^{-j\omega 2} + e^{-j\omega 4}] [-2 \text{sinc}(\omega) e^{j\omega} (1 - 3e^{-j6\omega})]$$

$$Z(\omega) = -2\text{sinc}(\omega)e^{-j\omega}(1 - 3e^{-j6\omega})(3 + e^{-j\omega 2})$$

(c)

Note: $x(t) = 2 \text{rect}(t/3)$ where $rect(t) = \begin{cases} 1, & |t| \leq 1.5 \\ 0, & |t| > 1.5 \end{cases}$

Then, the modulated signal can be given as

$$z(t) = x(t)\cos(\omega_0 t).$$

Taking the FT of $z(t)$,

$$Z(\omega) = F[z(t)] = F \left[x(t) \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) \right]$$

Based on the linearity property of the FT,

$$Z(\omega) = \frac{1}{2} F[x(t)e^{j\omega_0 t}] + \frac{1}{2} F[x(t)e^{-j\omega_0 t}]$$

Using the First-shifting theorem,

$$Z(\omega) = \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$

Note: $X(\omega) = F[\text{rect}(t/3)] = 3 \text{sinc}(\frac{\omega 3}{2})$.

So, finally we have

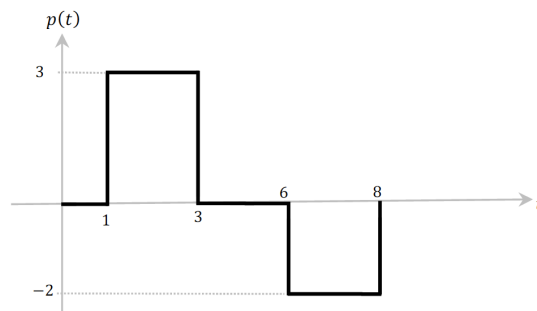
$$Z(\omega) = \frac{3}{2} \text{sinc}(1.5(\omega - \omega_0)) + \frac{3}{2} \text{sinc}(1.5(\omega + \omega_0))$$

(2020 Stage 2 Examinations, question 6)

(a)

[7 Marks]

An aperiodic continuous-time signal $p(t)$ is illustrated below. Is $p(t)$ a power signal or an energy signal? Calculate its power/energy as appropriate.



(b)

[10 Marks]

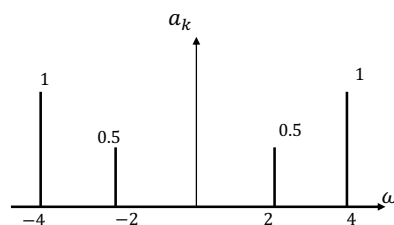
(i) Quoting which standard Fourier Transform pairs that you use, obtain the Fourier Transform $P(\omega)$ of the signal $p(t)$ given in part 6(a).

(ii) Also sketch the magnitude $|P(\omega)|$ over the frequency range $-3\pi \leq \omega \leq 3\pi$. Label all points of contact with either axis in your plot.

(c)

[8 Marks]

The Fourier Transform of a signal $x(t)$ is illustrated below. Determine an expression for $x(t)$.



SOLUTION

(a)

As $p(t)$ is not a periodic signal, this is not a power signal.

Also, $p(t)$ can be stated as an energy signal if its energy has a finite value, i.e., $\int_{-\infty}^{\infty} |p(\tau)|^2 d\tau < \infty$.

In particular, we can compute

$$E_p = \int_0^{\infty} |p(\tau)|^2 d\tau = \int_0^3 3^2 d\tau + \int_6^8 (-2)^2 d\tau = 18 + 8 = 26 < \infty$$

Thus, $p(t)$ is an energy signal with its energy value of 26.

(b)

(i)

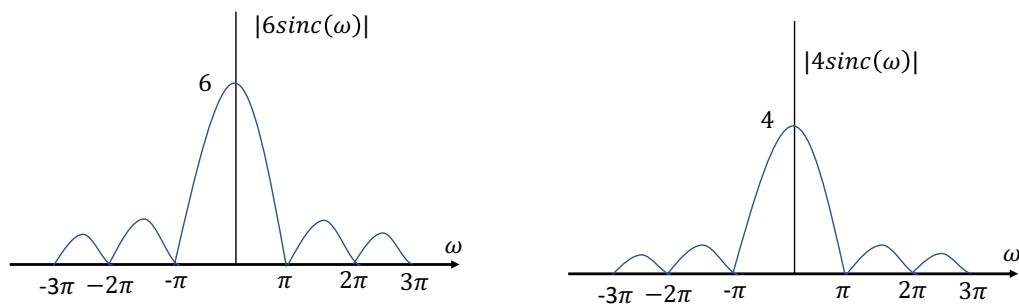
$$p(t) = 3 \operatorname{rect}\left(\frac{t-2}{2}\right) - 2 \operatorname{rect}\left(\frac{t-7}{2}\right)$$

This is a superposition of scaled two signals in the form of $\operatorname{rect}\left(\frac{t}{\tau}\right)$.

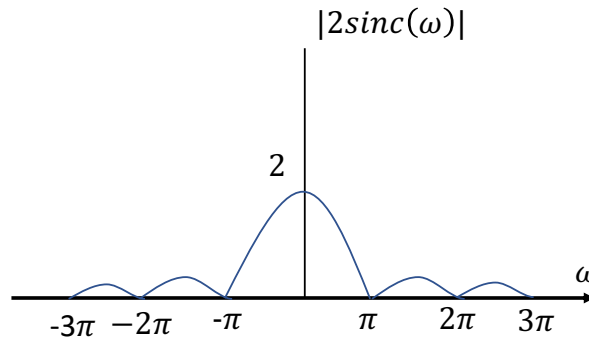
Note: $F\left[\operatorname{rect}\left(\frac{t}{\tau}\right)\right] = \tau \operatorname{sinc}\left(\omega \frac{\tau}{2}\right)$ and the Second-shifting theorem: $F[x(t-\tau)] = e^{-j\omega\tau}X(\omega)$.

$$\begin{aligned} P(\omega) &= 3 F\left[\operatorname{rect}\left(\frac{t-2}{2}\right)\right] - 2 F\left[\operatorname{rect}\left(\frac{t-7}{2}\right)\right] \\ &\Rightarrow P(\omega) = 3 e^{-j\omega 2} X_1(\omega) - 2 e^{-j\omega 7} X_2(\omega) \\ &\Rightarrow P(\omega) = 6 \operatorname{sinc}(\omega) e^{-j\omega 2} - 4 \operatorname{sinc}(\omega) e^{-j\omega 7} \end{aligned}$$

(ii)



Subtracting the right magnitude from the left magnitude spectrum, we have



Note that $e^{-j\omega\alpha}$ does not affect a magnitude but a phase only.

(c)

The Fourier Transform can be viewed as a superposition of two cosine spectra: one with amplitude 1 at $\omega = 2$ rad/s and one with amplitude 2 at $\omega = 4$ rad/s.

Thus, $X(\omega) = X1(\omega) + X2(\omega)$ where

$$X1(\omega) + X2(\omega) = \frac{1}{2}(\delta(\omega - 2) + \delta(\omega + 2)) + (\delta(\omega - 4) + \delta(\omega + 4))$$

Then, $x(t) = F^{-1}[X(\omega)] = F^{-1}[X1(\omega)] + F^{-1}[X2(\omega)]$, due to the linearity property of the FT. Accordingly, using the standard trigonometric FT pair of $F[\cos(\omega_0 t)] = \frac{1}{2}(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))$, we can obtain

$$x(t) = \cos(2t) + 2\cos(4t)$$

(2021 Stage 2 Examinations, question 3)

Two signals $x_1(t) = \sin(4t)$ and $x_2(t) = \cos(40t)$ are multiplied together to produce a signal $x(t) = x_1(t) \times x_2(t)$, as illustrated in Figure 3.

(a)

[8 Marks]

Use Fourier analysis to compute the average power of $x(t)$.

(b)

[4 Marks]

Compute the energy of the signal $x(t)$ in Part (a). Classify if this signal is a power signal or energy signal and explain your answer.

(c)

[8 Marks]

Sketch both the magnitude spectrum and the phase spectrum for $x(t)$, by using complex exponential Fourier series.

Label the values of all amplitudes $|c(\omega)|$ and phases $\angle c(\omega)$ in your plot.

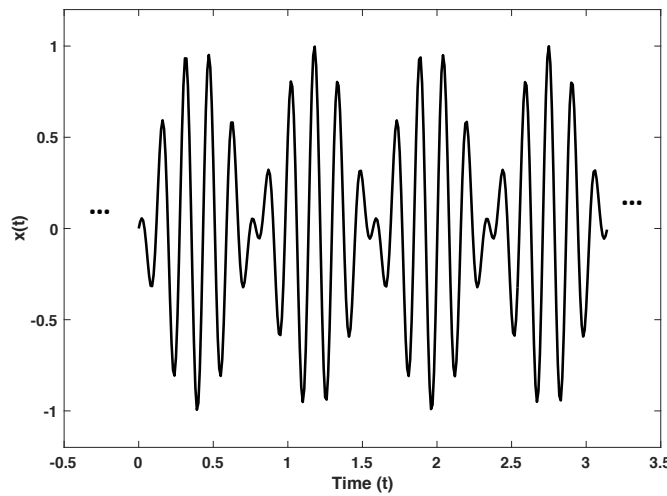


Figure 3. $x(t) = \sin(4t) \times \cos(40t)$

Solution

(a)

Using the Parseval's theorem, the average power of a period signal $x(t) = x_1(t) \times x_2(t)$ can be obtained as

$$P = \frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |c_n|^2 = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} |A_n|^2$$

where c_n are exponential Fourier coefficients and a_0 and $A_n = \sqrt{a_n^2 + b_n^2}$ are harmonic Fourier coefficients.

Thus, we can express $x(t)$ in the FS form by setting $\omega_0 = 4$ radian/sec, as

$$x(t) = \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t}) \times \frac{1}{2} (e^{j10\omega_0 t} + e^{-j10\omega_0 t})$$

$$x(t) = \frac{1}{4j} (-e^{-j11\omega_0 t} + e^{-j9\omega_0 t} - e^{j9\omega_0 t} + e^{j11\omega_0 t})$$

Thus, we have

$$P = \left| \frac{1}{4j} \right|^2 \times 4 = \frac{1}{16} \times 4 = \frac{1}{4}$$

where $c'_n s = \pm \frac{1}{4j}$ at $\omega = \pm 9\omega_0$ radian/sec and $\omega = \pm 11\omega_0$ radian/sec.

(b)

It is a power signal because the average power $P = \sum_{n=1}^{\infty} |c_n|^2 = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} |A_n|^2 = \frac{1}{4} < \infty$ is finite,

and the total energy of the signal $E_T = \int_{-\infty}^{\infty} |x(t)|^2 dt = \infty$ is infinite.

(c)

Note that in a complex exponential Fourier series form, $x(t)$ can be expressed as

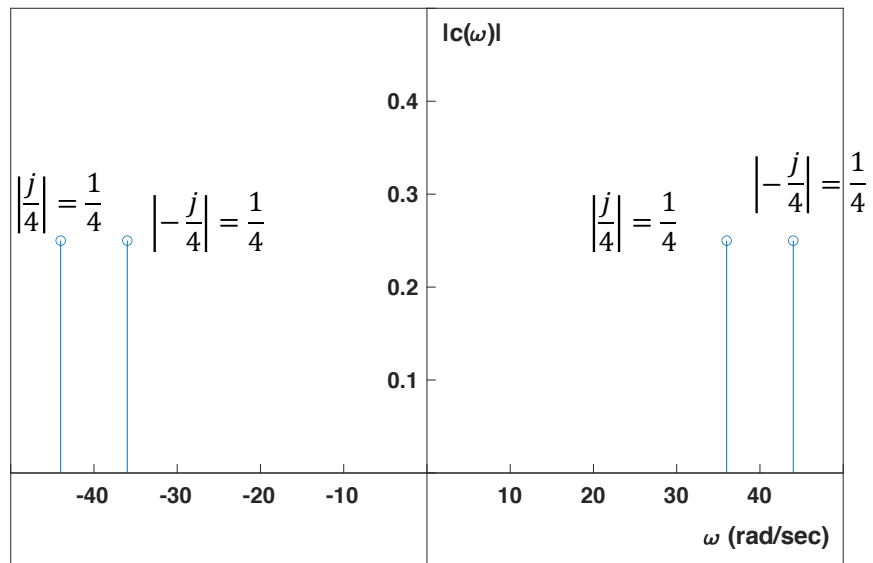
$$x(t) = \frac{1}{4j} (-e^{-j11\omega_0 t} + e^{-j9\omega_0 t} - e^{j9\omega_0 t} + e^{j11\omega_0 t})$$

where $\omega_0 = 4$ radian/sec.

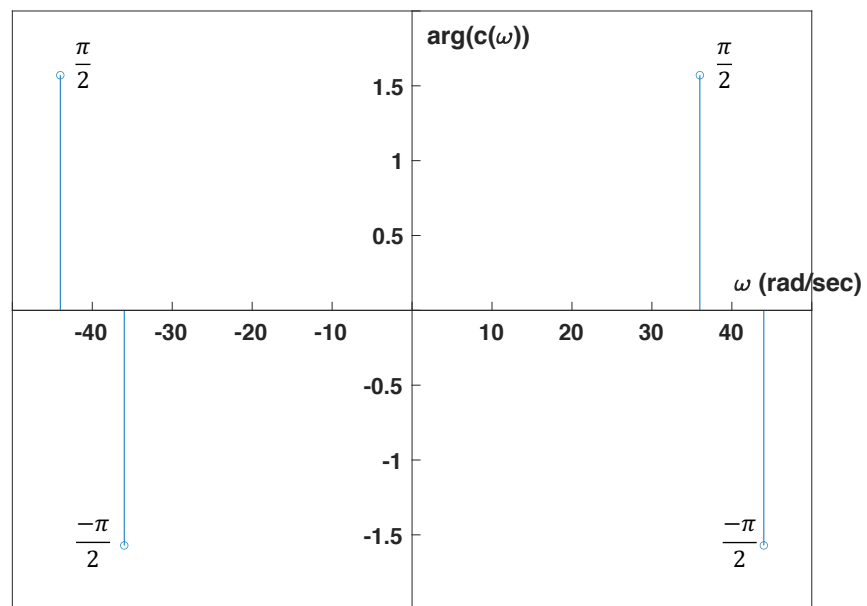
From this, we can observe that the Fourier coefficients c_n exist only at $\omega = -44, -36, 36$ and 44 radian/sec.

Accordingly, the magnitudes $|c(\omega)|$ and the phases $\angle c(\omega)$ are plotted in the following.

The magnitude spectrum:



The phase spectrum:



(2021 Stage 2 Examinations, question 4)

An aperiodic rectangular pulse $x(t) = \text{rect}\left(\frac{t}{4}\right)$ is applied to a linear time invariant system that has an impulse response $h(t) = \frac{1}{2}e^{j\omega_0 t}$, for given ω_0 .

- (a) [10 Marks]
Use the Fourier transform relation $\text{rect}\left(\frac{t}{\tau}\right) \leftrightarrow \tau \text{sinc}\left(\frac{\omega}{2}\tau\right)$ to determine the Fourier transform $Y(\omega)$ of the output signal $y(t) = h(t) * x(t)$, where the symbol $*$ denotes the convolution operation.

- (b) [4 Marks]
Determine $Z(\omega) = X(\omega) * [\delta(\omega - \pi) + \delta(\omega + \pi)]$, where $X(\omega)$ denotes the Fourier transform of $x(t)$ above and the symbol $*$ denotes the convolution operation.

- (c) [6 Marks]
When $x(t)$ is shifted by 6 in the time domain, i.e., $g(t) = x(t - 6)$, determine the Fourier transform of $x(t) + g(t)$, by using the provided tables of Fourier transform formulae.

Solution

(a)

The convolution integral in the time domain is given by

$$\begin{aligned} y(t) &= h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau \\ &= \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \end{aligned}$$

, as the output of the linear time invariant system, where its impulse response is $h(t)$ and an input is $x(t)$.

Applying the Fourier transform (FT) to the above convolution integral, we can have

$$\mathcal{F}[h(t) * x(t)] = \mathcal{F}[h(t)] \mathcal{F}[x(t)] = H(\omega)X(\omega)$$

where $X(\omega)$ and $H(\omega)$ denotes the FT of $x(t)$ and $h(t)$, respectively.

Noting that with $\tau = 4$,

$$\mathcal{F}\left[\text{rect}\left(\frac{t}{\tau}\right)\right] = \tau \text{sinc}\left(\frac{\omega}{2}\tau\right)$$

we can have:

$$X(\omega) = \mathcal{F} \left[x(t) = \text{rect} \left(\frac{t}{\tau} \right) \right] = 4 \text{sinc}(\omega 2)$$

Also, the FT of the impulse response will provide

$$H(\omega) = \mathcal{F}[h(t)] = \mathcal{F} \left[\frac{1}{2} e^{j\omega_0 t} \right] = \frac{1}{2} \delta(\omega - \omega_0)$$

by using the first shifting property:

$$F(\omega - \omega_1) \leftrightarrow f(t) e^{-j\omega_1 t}$$

and the FT of a delta function is

$$\delta(\omega) \leftrightarrow 1$$

Therefore, noting $H(\omega)X(\omega)$,

We can obtain

$$Y(\omega) = H(\omega)X(\omega) = \frac{1}{2} \delta(\omega - \omega_0) X(\omega) = 2 \delta(\omega - \omega_0) \text{sinc}(2\omega)$$

(b)

From the multiplicative property,

$$x_1(t)x_2(t) \leftrightarrow \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$$

So, note that

$$\begin{aligned} 2\pi \left(\frac{1}{2\pi} X(\omega) * \delta(\omega - \omega_c) \right) &\leftrightarrow 2\pi x(t) e^{j\omega_c t} \\ 2\pi \frac{1}{2\pi} X(\omega) * \delta(\omega + \omega_c) &\leftrightarrow 2\pi x(t) e^{-j\omega_c t} \end{aligned}$$

Accordingly, we can have

$$\begin{aligned} Z(\omega) &= 2\pi \mathcal{F}(x(t)e^{j\omega_c t} + x(t)e^{-j\omega_c t}) \\ \Rightarrow Z(\omega) &= 2\pi \mathcal{F}(x(t)e^{j\omega_c t}) + 2\pi \mathcal{F}(x(t)e^{-j\omega_c t}) \\ \Rightarrow Z(\omega) &= 2\pi X(\omega - \omega_c) + 2\pi X(\omega + \omega_c) \end{aligned}$$

With $\omega_c = \pi$, note that $Z(\omega)$ is the scaled, shifted version of $X(\omega) = 4 \text{sinc}(2\omega)$ by $\omega_c = \pm\pi$ rad/sec in the frequency domain. Thus, $Z(\omega)$ is obtained in the following:

$$Z(\omega) = 8\pi \left(\text{sinc}(2(\omega - \pi)) + \text{sinc}(2(\omega + \pi)) \right)$$

(c)

The signal is a superposition of two aperiodic signals where

$$x(t) = \text{rect}\left(\frac{t}{\tau}\right)$$

and

$$g(t) = \text{rect}\left(\frac{t-6}{\tau}\right)$$

Taking $\tau = 4$, $X(\omega)$ is given by

$$X(\omega) = 4\text{sinc}(\omega 2) = \frac{2 \sin(2\omega)}{\omega}$$

where $\text{sinc}(a) = \frac{\sin(a)}{a}$.

And accounting for the delay on $x(t)$: $g(t) = x(t-4)$, and referring to the second-shifting theorem of

$$f(t-t_1) \leftrightarrow F(\omega)e^{-j\omega t_1}$$

We can have

$$G(\omega) = e^{-j6\omega}X(\omega) = e^{-j6\omega}\frac{2 \sin(2\omega)}{\omega}$$

Hence, applying the linearity property of the Fourier transform, the FT of the superposition is

$$\begin{aligned} X(\omega) + G(\omega) &= \frac{2 \sin(2\omega)}{\omega} + \frac{e^{-6j\omega} 2 \sin(2\omega)}{\omega} \\ &= \frac{2 \sin(2\omega)}{\omega} (1 + e^{-6j\omega}) = 4 \text{sinc}(2\omega)(1 + e^{-j6\omega}) \end{aligned}$$

(2022 Stage 2 Examinations, question 3)

(a) [8 marks]

Derive the frequency-domain convolution of a spectrum $M(\omega) = \text{rect}\left(\frac{\omega}{B}\right)$ and a spectrum $G(\omega) = \delta(\omega - 10) + \delta(\omega + 10)$, where $B > 0$ and $\text{rect}\left(\frac{\omega}{B}\right)$ is a rectangular spectrum with width B . Also, sketch the spectrum of the convolution output.

(b) [12 marks]

Derive the inverse Fourier transform of $M(\omega) * G(\omega)$ in part Q3a, where the symbol $*$ denotes the convolution in the frequency domain. Quote which transform pairs that you use.

Solution

(a)

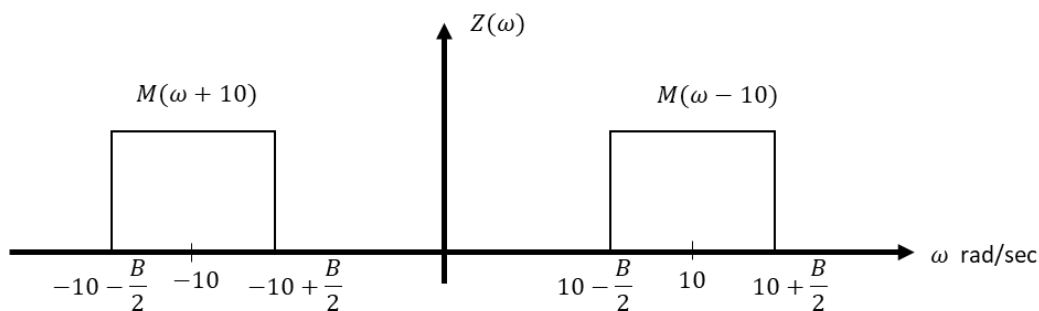
Using the convolution in the frequency domain,

$$\begin{aligned} Z(\omega) &= M(\omega) * G(\omega) = \int_{-\infty}^{\infty} G(\tau) M(\omega - \tau) d\tau \\ &= \int_{-\infty}^{\infty} [\delta(\tau - 10) + \delta(\tau + 10)] M(\omega - \tau) d\tau \\ &= M(\omega - 10) + M(\omega + 10) \end{aligned}$$

Thus, we have that $Z(\omega)$ is the superposition of the shifted versions of $M(\omega)$:

$$Z(\omega) = \text{rect}\left(\frac{\omega - 10}{B}\right) + \text{rect}\left(\frac{\omega + 10}{B}\right)$$

Sketch the spectrum of $Z(\omega)$: The spectrum of $Z(\omega)$ contains $M(\omega)$ shifted to both $\omega = 10$ rad/sec and $\omega = -10$ rad/sec.



(b)

One possible approach is to start from using the convolution integral in the frequency domain, which results in the following:

$$Z(\omega) = M(\omega) * G(\omega) = M(\omega - 10) + M(\omega + 10)$$

Using the linearity and the First shifting theorem that

$$X(\omega - \omega_0) = F[x(t)e^{j\omega_0 t}]$$

we can have

$$\begin{aligned} z(t) &= F^{-1}(Z(\omega)) = m(t)e^{j10t} + m(t)e^{-j10t} \\ z(t) &= 2m(t) \cos(10t) \end{aligned}$$

Here, note that we can obtain

$$m(t) = F^{-1}\left[rect\left(\frac{\omega}{B}\right)\right]$$

by applying the duality property that

$$2\pi x(-\omega) \leftrightarrow X(t)$$

to the standard result that

$$rect\left(\frac{t}{\tau}\right) \leftrightarrow \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$

Thus, we have

$$m(t) = \frac{1}{2\pi} B \operatorname{sinc}\left(t \frac{B}{2}\right)$$

Finally, we can express for $z(t)$ as

$$z(t) = \frac{1}{\pi} B \operatorname{sinc}\left(t \frac{B}{2}\right) \cos(10t).$$

Or,

$$z(t) = \frac{1}{2\pi} B \operatorname{sinc}\left(t \frac{B}{2}\right) (e^{j10t} + e^{-j10t})$$

(2022 Stage 2 Examinations, question 4)

(a) [8 marks]

Use Fourier analysis to compute the average power of a signal $x(t) = 4 \cos(5t) \sin(10t)$.

(b) [12 marks]

Derive the Fourier spectrum of a signal that is given by

$$z(t) = \int_{-\infty}^{\infty} \text{rect}(2\tau) \exp(-j10\tau + jt) d\tau.$$

Also, sketch the magnitude spectrum of $z(t)$.

Solution

(a)

The average power of a periodic signal can be obtained as

$$P = \frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt$$

where T is one periodic interval.

Using the Parseval's theorem, the average power of the periodic signal over $\langle T \rangle$ can be given by

$$P = \sum_n |c_n|^2 = a_0^2 + \frac{1}{2} \sum_n |A_n|^2$$

Note that using

$$\cos(5t) = \frac{1}{2} (e^{j5t} + e^{-j5t})$$

$$\sin(10t) = \frac{1}{2j} (e^{j10t} - e^{-j10t})$$

we can express in a complex exponential form of the Fourier Series

$$\begin{aligned} x(t) &= 4 \cos(5t) \sin(10t) = 4 \frac{1}{2} (e^{j5t} + e^{-j5t}) \frac{1}{2j} (e^{j10t} - e^{-j10t}) \\ &= \frac{1}{j} (e^{j15t} - e^{-j5t} + e^{j5t} - e^{-j15t}) \end{aligned}$$

Thus, the average power of the given signal $x(t)$ is

$$P = \sum |c_n|^2 = \left| \frac{1}{j} \right|^2 \times 4 = 4,$$

where $\omega_0 = 5$ rad/sec and the complex exponential Fourier coefficients are specified as $c_1 = \frac{1}{j}$, $c_{-1} = -\frac{1}{j}$, $c_3 = \frac{1}{j}$, and $c_{-3} = -\frac{1}{j}$

(b)

In the form of the convolution integral, we can re-express the given signal in the following:

$$\begin{aligned} z(t) &= \int_{-\infty}^{\infty} \text{rect}(2\tau) e^{-j9\tau} e^{-j\tau} e^{jt} d\tau \\ &= \int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau = x(t) * y(t) \end{aligned}$$

Where $x(t) = \text{rect}(2t) e^{-j9t}$ and $y(t) = e^{jt}$.

Then, using the time convolution property of the Fourier transform, we can express that

$$Z(\omega) = F[z(t) = x(t) * y(t)] = F[x(t)] \times F[y(t)]$$

Firstly, we can obtain $F[y(t)] = \delta(\omega - 1)$ from the standard results that $F[e^{j\omega_0 t}] = \delta(\omega - \omega_0)$ and $F[1] = \delta(\omega)$.

Secondly, using the First shifting theorem that

$$F[g(t)e^{j\omega_0 t}] = G(\omega - \omega_0)$$

and the standard result that

$$F\left[\text{rect}\left(\frac{t}{\tau}\right)\right] = \tau \text{sinc}\left(\omega \frac{\tau}{2}\right)$$

we can obtain

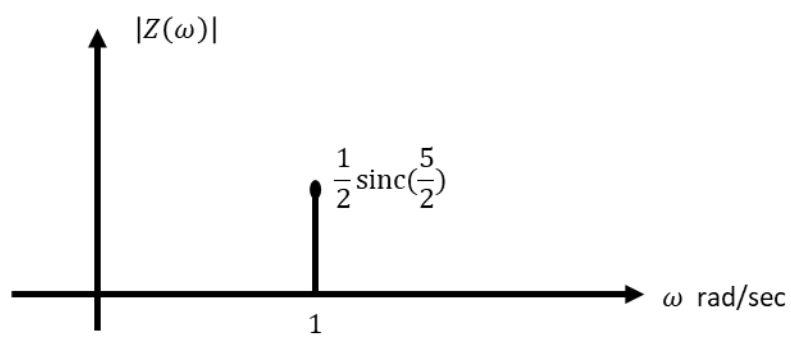
$$F[x(t)] = F[g(t)e^{-j9t}] = G(\omega + 9) = \frac{1}{2} \text{sinc}\left(\frac{\omega + 9}{4}\right)$$

where $g(t) = \text{rect}(2t)$.

Finally, the Fourier spectrum of $z(t)$ is

$$Z(\omega) = \frac{1}{2} \text{sinc}\left(\frac{\omega + 9}{4}\right) \delta(\omega - 1)$$

To sketch the magnitude spectrum of $Z(\omega)$, notice that $\delta(\omega - 1)$ returns zero except at $\omega = 1$ rad/sec and thus, it produces the magnitude spectrum as



(2023 Stage 2 Examinations, question 3)

(a) [10 Marks]

Using the exponential Fourier series expression, determine the Fourier series spectrum (Line spectrum) of the following signal:

$$x(t) = \sum_{n=-\infty}^{\infty} p(t - 2n)$$

where $p(t) = 1$ for $0 \leq t \leq 1$ and zero for $1 < t < 2$.

Sketch the Line spectrum over the frequency range $-3\pi \leq \omega \leq 3\pi$ (rad/sec). Label your axes carefully in your plot.

(b)

[10 Marks]

Determine the Fourier transform $X(\omega)$ of the aperiodic signal $x(t)$, which is given by

$$x(t) = \cos(10t) e^{-|t|}$$

State which standard properties of the Fourier transform you have used.

Solution

(a)

For a periodic signal $x(t)$, we can express it with the exponential Fourier series form:

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

Where C_n are exponential Fourier coefficients and $\omega_0 = 2\pi/T$ with $T = 2$

We can find

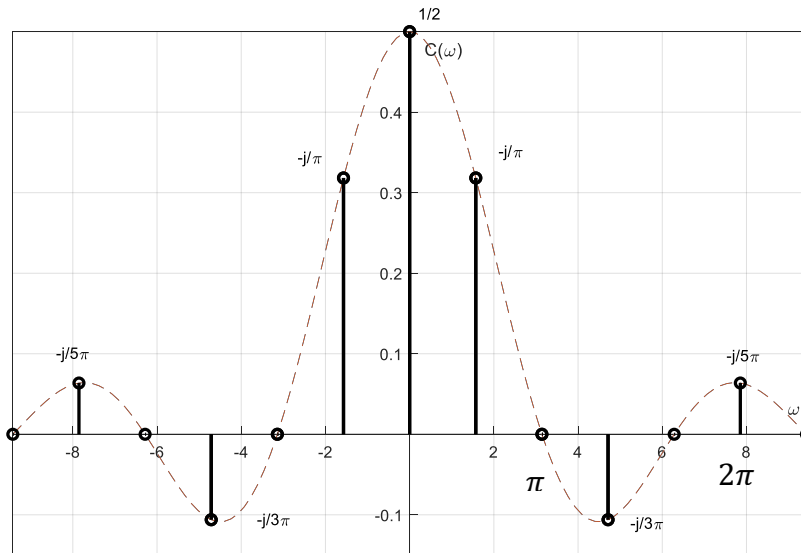
$$\begin{aligned} C_n &= \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jn\omega_0 t} dt \\ C_n &= \frac{1}{2} \int_0^2 x(t) e^{-jn\omega_0 t} dt = \frac{1}{2} \int_0^1 e^{-jn\omega_0 t} dt \\ C_n &= \frac{1}{n\pi} e^{-\frac{jn\pi}{2}} \sin\left(\frac{n\pi}{2}\right) = \frac{1}{2} \text{sinc}\left(\frac{n\pi}{2}\right) e^{-\frac{jn\pi}{2}} \end{aligned}$$

Accordingly, the Line spectrum of $x(t)$ in terms of C_n is

$$C(\omega) = \sum_n \frac{1}{2} \text{sinc}\left(\frac{n\pi}{2}\right) 2\pi \delta\left(\omega - \frac{n\pi}{2}\right) = \sum_n \pi \text{sinc}\left(\frac{n\pi}{2}\right) \delta\left(\omega - \frac{n\pi}{2}\right)$$

Or if using $F[e^{-j\tilde{\omega}t}] = \delta(\omega + \tilde{\omega})$

$$C(\omega) = \sum_n \frac{1}{2} \text{sinc}\left(\frac{n\pi}{2}\right) \delta\left(\omega - \frac{n\pi}{2}\right)$$



Answer to Question 3

(b)

We can consider that

$$x(t) = \cos(10t) \{x_1(t) + x_1(-t)\}$$

where $x_1(t) = e^{-t}u(t)$ and $x_1(-t) = e^t u(-t)$

Then, using the linearity of the Fourier transform, we can have

$$X(\omega) = F[x(t)] = F[\cos(10t) x_1(t)] + F[\cos(10t) x_1(-t)]$$

Here, we have

$$\begin{aligned} \int_0^\infty x_1(t) e^{-j\omega t} dt + \int_{-\infty}^0 x_1(-t) e^{-j\omega t} dt \\ \Rightarrow X_1(\omega) + X_1(-\omega) \end{aligned}$$

when using $X_1(\omega) = F[x_1(t)]$ and the time reversal property of the Fourier transform:

$F[y(-t)] = Y(-\omega)$ for a given $y(t)$

Note that,

$$X_1(\omega) = \int_0^{\infty} e^{-t} e^{-j\omega t} dt = \frac{1}{1+j\omega}$$

and,

$$X_1(-\omega) = \int_0^{\infty} e^{-t} e^{j\omega t} dt = \frac{1}{1-j\omega}$$

To obtain $F[\cos(10t) x_1(t)]$ in $X(\omega)$, we have

$$\int_0^{\infty} \cos(10t) e^{-t} e^{-j\omega t} dt = \frac{1}{2} (X_1(\omega - 10) + X_1(\omega + 10))$$

by using the standard Fourier transform pairs of

$$F[y(t) \cos(\omega_0 t)] = \frac{1}{2} Y(\omega - \omega_0) + \frac{1}{2} Y(\omega + \omega_0)$$

Similarly, we have

$$F[\cos(10t) x_1(-t)] = \frac{1}{2} X_1(-(\omega - 10)) + \frac{1}{2} X_1(-(\omega + 10))$$

Using these, we have

$$\begin{aligned} \frac{1}{2} (X_1(\omega - 10) + X_1(\omega + 10)) &= \frac{1}{2} \left(\frac{1}{1+j(\omega - 10)} + \frac{1}{1+j(\omega + 10)} \right) \\ \frac{1}{2} X_1(-(\omega - 10)) + \frac{1}{2} X_1(-(\omega + 10)) &= \frac{1/2}{1-j(\omega - 10)} + \frac{1/2}{1-j(\omega + 10)} \end{aligned}$$

Finally, we provide

$$\begin{aligned} X(\omega) &= \frac{1}{2} \left[\frac{1}{1+j(\omega - 10)} + \frac{1}{1+j(\omega + 10)} + \frac{1}{1-j(\omega - 10)} + \frac{1}{1-j(\omega + 10)} \right] \\ &= \frac{2}{1 + (\omega - 10)^2} + \frac{2}{1 + (\omega + 10)^2} \end{aligned}$$

(2023 Stage 2 Examinations, question 4)

(a)

[8 Marks]

An aperiodic signal $x(t)$ is defined as

$$x(t) = \begin{cases} \cos(2\pi t), & -3 \leq t \leq 3 \\ 0, & \text{else} \end{cases}$$

Determine whether $x(t)$ is a power signal or an energy signal. Calculate its power/energy as appropriate.

(b)

[12 Marks]

Determine the magnitude spectrum $|X(\omega)|$ of the signal $x(t)$ in Part (a) by quoting which standard Fourier transform pairs that you use.

Sketch $|X(\omega)|$ over the frequency range $\pi \leq |\omega| \leq 3\pi$ (rad/sec). Label your axes carefully in your plot.

Solution

(a)

As $x(t)$ is not periodic with a finite length, it is not a power signal. We can state that $x(t)$ is an energy signal if its energy has a finite value, i.e.,

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega < \infty$$

To determine if it is an energy signal, we may consider either $x(t)$ or $X(\omega)$.

But, in this case, $X(\omega)$ has an infinite length as a sinc function. Thus, we need to compute its total energy with $x(t)$.

$$\begin{aligned} E_{total} &= \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-3}^3 \frac{1}{4} (e^{j2\pi t} + e^{-j2\pi t})^2 dt \\ &= \frac{1}{4} \left[\frac{1}{j4\pi} (e^{j12\pi} - e^{-j12\pi}) - \frac{1}{j4\pi} (e^{-j12\pi} - e^{j12\pi}) + 12 \right] \\ &= \frac{1}{2} \frac{1}{j4\pi} (e^{j12\pi} - e^{-j12\pi}) + 3 \\ &= \frac{1}{4\pi} \sin(12\pi) + 3 = 3. \end{aligned}$$

(** alternatively, use the identity $\cos^2(\theta) = (\cos(2\theta) + 1)/2$ and we can have $E_{Total} = \frac{1}{2} \int_{-3}^3 (\cos(4\pi t) + 1) dt = 3$ ***)

Thus, we can state that this signal is an energy signal with $E_{Total} = 3$

(b)

Note that $x(t)$ can be viewed as a modulation signal. That is, a rectangular pulse with unit height and width of 6 centred around at $t = 0$ is modulated by $\cos(2\pi t)$.

In this view, the Fourier transform spectrum $X(\omega)$ of this modulated signal is

$$X(\omega) = \frac{1}{2} X_r(\omega - 2\pi) + \frac{1}{2} X_r(\omega + 2\pi)$$

where we use the standard Fourier transform pair of

$$F[\cos(\omega_0) y(t)] = \frac{1}{2} Y(\omega - \omega_0) + \frac{1}{2} Y(\omega + \omega_0)$$

with

$$X_r(\omega) = F \left[\text{rect} \left(\frac{t}{6} \right) \right]$$

and $\text{rect} \left(\frac{t}{6} \right) = u(t + 3) - u(t - 3)$ denotes the rectangular pulse with unit height and width of 6.

Using the standard Fourier transform pair of $F \left[\text{rect} \left(\frac{t}{T} \right) \right] = T \text{sinc} \left(\frac{\omega T}{2} \right)$,

$$X_r(\omega) = \frac{6 \sin(\omega 3)}{3\omega} = 6 \text{sinc}(3\omega).$$

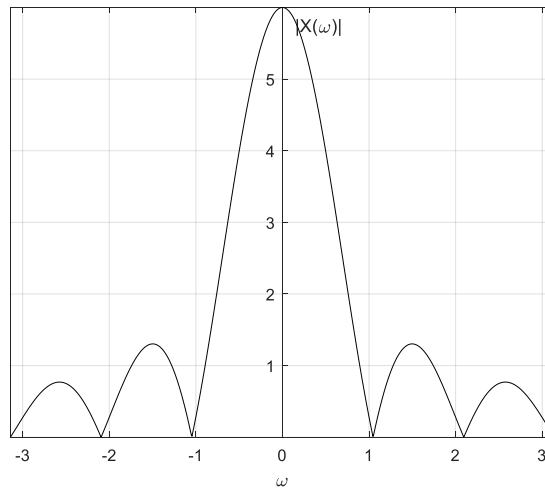
Based on these, we have

$$X(\omega) = 3 \text{sinc}(3(\omega - 2\pi)) + 3 \text{sinc}(3(\omega + 2\pi)).$$

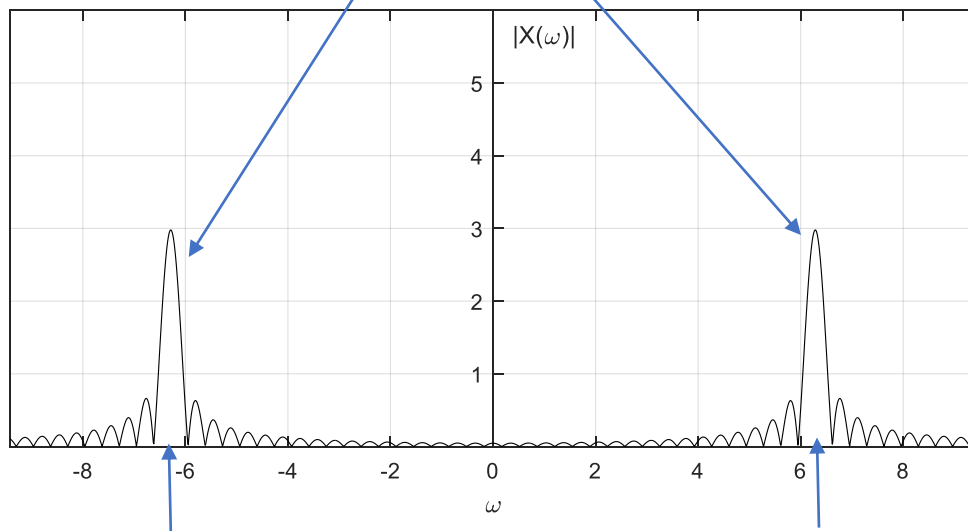
Therefore, the magnitude spectrum is given by

$$|X(\omega)| = 3 |\text{sinc}(3(\omega - 2\pi)) + \text{sinc}(3(\omega + 2\pi))|$$

$|X_r(\omega)|$ is sketched over the frequency range $|\omega| \leq \pi$ in the following.



The magnitude spectrum $|X(\omega)|$ is



Centred at $\omega = -2\pi$

Centred at $\omega = 2\pi$